# Mariel Del Rosario

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## EDUCATION

**University of California, Berkeley |** GPA: 3.9/4.0 | Graduated With General Distinction **Berkeley, CA**
Bachelor of Arts in Geography, Highest Honors *May 2026*

## TECHNICAL SKILLS

**Programming and Scripting:** Python (Pandas, NumPy, ArcPy, GeoPandas basics), R, SQL, HTML/CSS, JavaScript
**Data Analysis and Dev Tools:** Statistical Modeling (LPM, Hypothesis Testing), Jupyter/Colab, Git/GitHub, Excel, Google Sheets Automation, REST APIs
**Geospatial Technologies:** ArcGIS Pro, ArcGIS Online, ArcGIS StoryMaps, QGIS, GDAL, Spatial Analysis (Suitability, Network/Service Area, Interpolation, Geocoding), CRS Reprojection

## TECHNICAL PROJECTS

**MUNI Transit Suitability Analysis (ArcGIS Pro, Spatial Analysis)** **Fall 2025**
- Built a multi-criteria spatial suitability model in ArcGIS Pro to evaluate optimal public transit station locations across San Francisco
- Conducted Euclidean distance, network service area, Spline interpolation, and market share spatial analyses on multi-layered municipal datasets
- Published findings and interactive map visualizations via ArcGIS StoryMaps to communicate transit equity solutions for low-vehicle-ownership communities

**Native American Health Equity Spatial Analysis (QGIS, Public Data)** **Spring 2025**
- Ingested, cleaned, and normalized census and regional public health datasets to analyze socioeconomic disparities across Riverside County, CA
- Engineered 4 thematic map layers in QGIS visualizing correlations between the Health Equity Index, land use patterns, and localized health outcomes

**PEPFAR/DREAMS Quantitative Health Study (R, Statistical Modeling)** **Spring 2026**
- Conducted an independent 19-page quantitative study analyzing demographic and geographic health metrics across 44,000+ records from DHS surveys
- Developed R scripts to clean, recode, and validate 17 years of pooled cross-sectional spatial and survey data
- Implemented Linear Probability Models (LPM) and clustered standard error regressions to isolate program impact from regional demographic controls

## RESEARCH EXPERIENCE

**UC Berkeley Geography Department** **Berkeley, CA**
*Geospatial Research Assistant for Dr. Tianna Bruno* *August 2024 – Present*
- Georeference historic spatial assets and locations into interactive spatial databases and digital mapping platforms
- Design spatial data repositories, managing data organization and standardization for community-facing cartographic tools
- Co-authored and presented spatial environmental justice research at the 2026 AAG Annual Conference
- Collaborated in a 4-person team to build interactive digital storymaps documenting regional environmental injustice and historical geography

## WORK EXPERIENCE

**College of Computing, Data Science, and Society at UC Berkeley** **Berkeley, CA**
*Lead Teaching Assistant and Senior Mentor for Foundations of Data Science* *August 2023 – May 2026*
- Automated backend workflows for 34 staff members using Google Colab and Google Sheets integrations
- Co-authored, tested, and debugged Python programming assignments and labs for 1,500+ students
- Supervised 40+ TAs/tutors on technical pedagogy, Python debugging, and statistical inference

**Pilipinx Academic Student Services** **Berkeley, CA**
*Retention Director and Gender/Sexuality Awareness Coordinator* *May 2023 – June 2025*
- Developed custom spatial databases via Google Maps to optimize vendor routing, venue selection, and logistical event planning for large-scale operations
- Managed a $26,000+ budget and led a 15-person team organizing student retention initiatives